Animal Kingdom

4 Animal Kingdom

When you look around, you will observe different animals with different structures and forms. As over a **million species** of animals have been described till now, the need for classification becomes all the more important. The classification also helps in assigning a systematic position to newly described species. This chapter runs in two parts: **4.1 Basis of Classification** and **4.2 Classification of Animals**.

4.1 Basis of Classification

Inspite of differences in structure and form of different animals, there are fundamental features common to various individuals in relation to the arrangement of cells, body symmetry, nature of coelom, and patterns of digestive, circulatory or reproductive systems. These features are used as the basis of animal classification and some of them are discussed here.

4.1.1 Levels of Organisation

Though all members of Animalia are multicellular, all of them do not exhibit the same pattern of organisation of cells.

- **Cellular level:** in sponges, the cells are arranged as loose cell aggregates. Some division of labour (activities) occurs among the cells.
- **Tissue level:** in coelenterates, the arrangement of cells is more complex. Cells performing the same function are arranged into tissues.
- **Organ level:** exhibited by Platyhelminthes and other higher phyla, where tissues are grouped together to form organs, each specialised for a particular function.
- **Organ system level:** in Annelids, Arthropods, Molluscs, Echinoderms and Chordates, organs have associated to form functional systems, each system concerned with a specific physiological function.

Organ systems in different groups of animals exhibit various patterns of complexities. Two examples are the digestive and the circulatory systems:

- The **digestive system** in Platyhelminthes has only a single opening to the outside of the body that serves as both mouth and anus, and is hence called **incomplete**. A **complete** digestive system has two openings, mouth and anus.
- The **circulatory system** is of two types: (i) **open type** in which the blood is pumped out of the heart and the cells and tissues are directly bathed in it; (ii) **closed type** in which the blood is circulated through a series of vessels of varying diameters (arteries, veins and capillaries).

4.1.2 Symmetry

Animals can be categorised on the basis of their symmetry.

- **Asymmetry:** sponges are mostly asymmetrical, i.e., any plane that passes through the centre does not divide them into equal halves.
- **Radial symmetry:** when any plane passing through the central axis of the body divides the organism into two identical halves. Coelenterates, ctenophores and echinoderms have this kind of body plan (Figure 4.1a).
- **Bilateral symmetry:** animals like annelids, arthropods, etc., where the body can be divided into identical left and right halves in only one plane (Figure 4.1b).

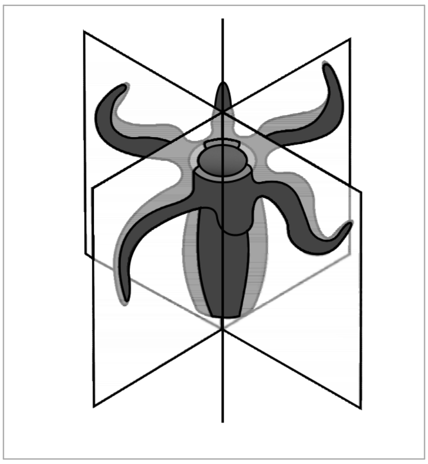

Fig. 4.1(a) - *Radial symmetry.*

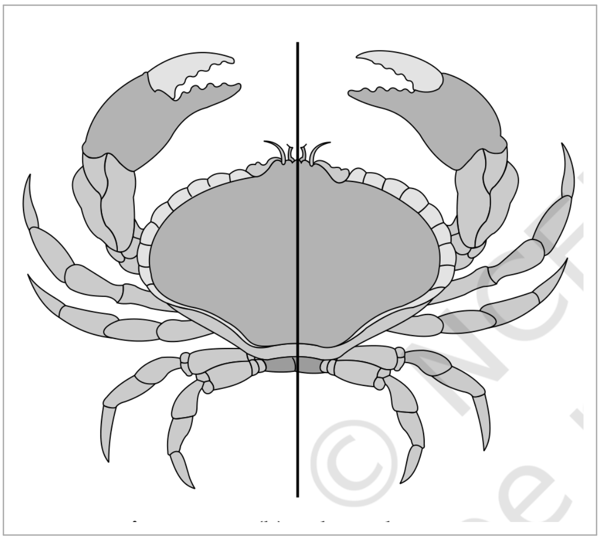

Fig. 4.1(b) - *Bilateral symmetry.*

4.1.3 Diploblastic and Triploblastic Organisation

- **Diploblastic animals:** those in which the cells are arranged in two embryonic layers, an external **ectoderm** and an internal **endoderm**, e.g., coelenterates. An undifferentiated layer, **mesoglea**, is present in between the ectoderm and the endoderm (Figure 4.2a).
- **Triploblastic animals:** those in which the developing embryo has a third germinal layer, **mesoderm**, in between the ectoderm and endoderm (platyhelminthes to chordates, Figure 4.2b).

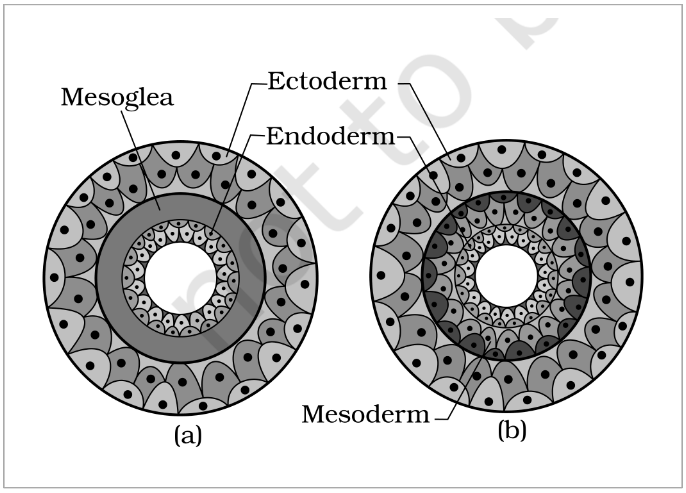

Fig. 4.2 - *Showing germinal layers : (a) Diploblastic (b) Triploblastic. Labels: Ectoderm, Mesoglea, Endoderm and Mesoderm.*

4.1.4 Coelom

Presence or absence of a cavity between the body wall and the gut wall is very important in classification. The body cavity, which is lined by mesoderm, is called **coelom**.

- **Coelomates:** animals possessing coelom, e.g., annelids, molluscs, arthropods, echinoderms, hemichordates and chordates (Figure 4.3a).
- **Pseudocoelomates:** in some animals the body cavity is not lined by mesoderm; instead the mesoderm is present as scattered pouches in between the ectoderm and endoderm. Such a body cavity is called **pseudocoelom**, e.g., aschelminthes (Figure 4.3b).
- **Acoelomates:** animals in which the body cavity is absent, e.g., platyhelminthes (Figure 4.3c).

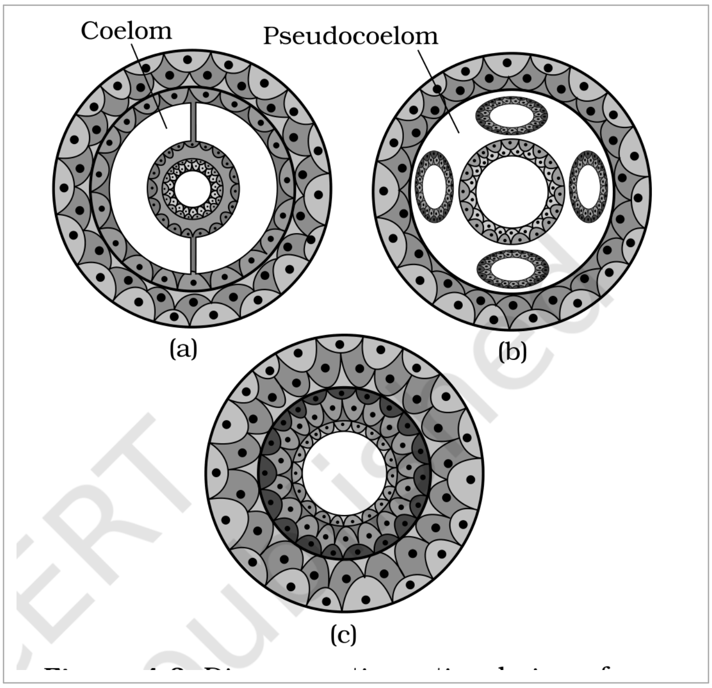

Fig. 4.3 *- Diagrammatic sectional view of : (a) Coelomate (b) Pseudocoelomate (c) Acoelomate. Labels: Coelom and Pseudocoelom.*

4.1.5 Segmentation

In some animals, the body is externally and internally divided into segments with a serial repetition of at least some organs. For example, in earthworm, the body shows this pattern called **metameric segmentation** and the phenomenon is known as **metamerism**.

4.1.6 Notochord

Notochord is a mesodermally derived rod-like structure formed on the dorsal side during embryonic development in some animals. Animals with notochord are called **chordates** and those animals which do not form this structure are called **non-chordates**, e.g., porifera to echinoderms.

4.2 Classification of Animals

The broad classification of Animalia, based on common fundamental features as mentioned in the preceding sections, is given in Figure 4.4. The important characteristic features of the different phyla are described in the sections that follow.

- The classification chart branches the **Kingdom Animalia (multicellular)** by **Levels of Organisation**, then by **Symmetry**, and then by **Body Cavity or Coelom**, ending in the **Phylum** leaves. From the cellular level branch come **Porifera** (mostly asymmetrical). From the **Tissue/Organ/Organ system** levels branch, by **Radial** symmetry, come **Coelenterata (Cnidaria)** and **Ctenophora**.
- Under **Bilateral** symmetry the animals are grouped by body cavity: **Without body cavity (acoelomates)** gives **Platyhelminthes**; **With false coelom (pseudocoelomates)** gives **Aschelminthes**; and **With true coelom (coelomates)** gives **Annelida**, **Arthropoda**, **Mollusca**, **Echinodermata**, **Hemichordata** and **Chordata**.

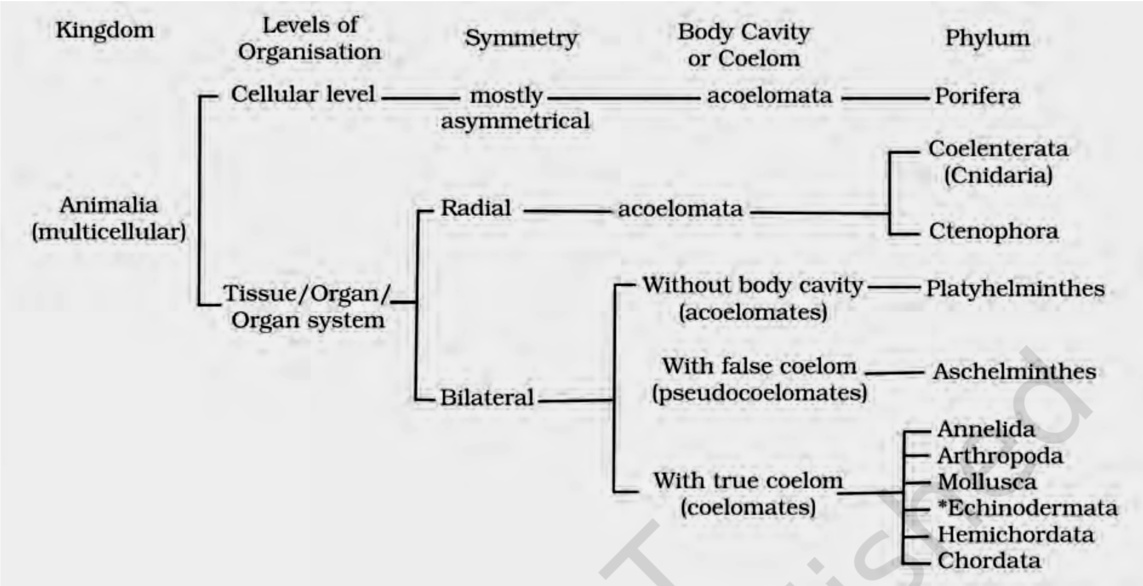

Fig. 4.4 *- Broad classification of Kingdom Animalia based on common fundamental features.*

> ⓘ ***[NOTE]*** *Echinodermata exhibits radial or bilateral symmetry depending on the stage.*

4.2.1 Phylum - Porifera

- Members of this phylum are commonly known as **sponges**. They are generally marine and mostly asymmetrical animals (Figure 4.5). These are primitive multicellular animals and have **cellular level** of organisation.
- Sponges have a **water transport or canal system**. This pathway of water transport is helpful in food gathering, respiratory exchange and removal of waste.

1. Water enters through minute pores (**ostia**) in the body wall.
2. It passes into a central cavity, the **spongocoel**.
3. Water goes out through the **osculum**.

- **Choanocytes** or collar cells, which are flagellated, line the spongocoel and the canals. Digestion is **intracellular**. The body is supported by a skeleton made up of **spicules** or **spongin fibres**.
- Sexes are not separate (**hermaphrodite**), i.e., eggs and sperms are produced by the same individual. Sponges reproduce asexually by fragmentation and sexually by formation of gametes. Fertilisation is internal and development is indirect, having a larval stage which is morphologically distinct from the adult.
- **Examples:** *Sycon* (Scypha), *Spongilla* (Fresh water sponge) and *Euspongia* (Bath sponge).

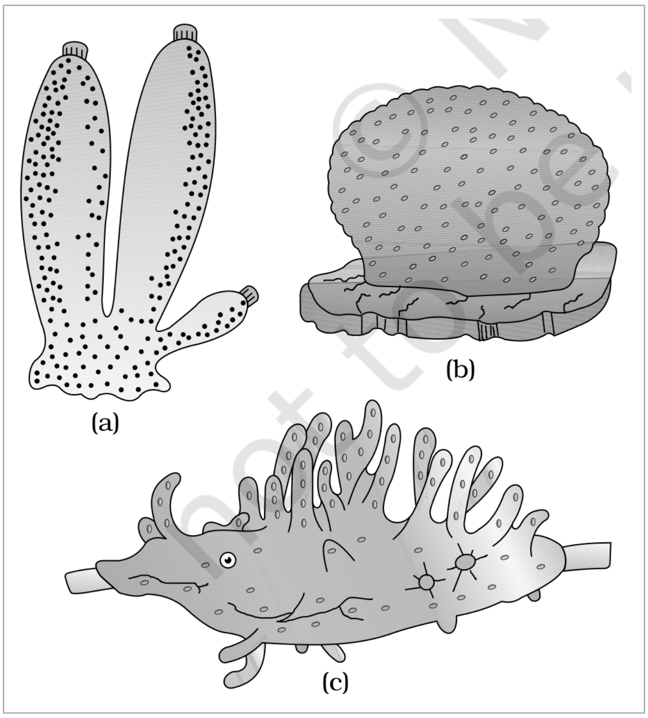

Fig. 4.5 *- Examples of Porifera : (a) Sycon (b) Euspongia (c) Spongilla.*

4.2.2 Phylum - Coelenterata (Cnidaria)

- They are aquatic, mostly marine, sessile or free-swimming, radially symmetrical animals (Figure 4.6). The name cnidaria is derived from the **cnidoblasts** or cnidocytes (which contain the stinging capsules or **nematocysts**) present on the tentacles and the body. Cnidoblasts are used for anchorage, defense and for the capture of prey (Figure 4.7).
- Cnidarians exhibit **tissue level** of organisation and are **diploblastic**. They have a central **gastro-vascular cavity** with a single opening, mouth on hypostome. Digestion is extracellular and intracellular. Some cnidarians, e.g., corals, have a skeleton composed of calcium carbonate.
- Cnidarians exhibit two basic body forms called **polyp** and **medusa** (Figure 4.6). The former is a sessile and cylindrical form like *Hydra*, *Adamsia*, etc., whereas the latter is umbrella-shaped and free-swimming like *Aurelia* or jelly fish.
- Those cnidarians which exist in both forms exhibit **alternation of generations (Metagenesis)**, i.e., polyps produce medusae asexually and medusae form the polyps sexually (e.g., *Obelia*).
- **Examples:** *Physalia* (Portuguese man-of-war), *Adamsia* (Sea anemone), *Pennatula* (Sea-pen), *Gorgonia* (Sea-fan) and *Meandrina* (Brain coral).

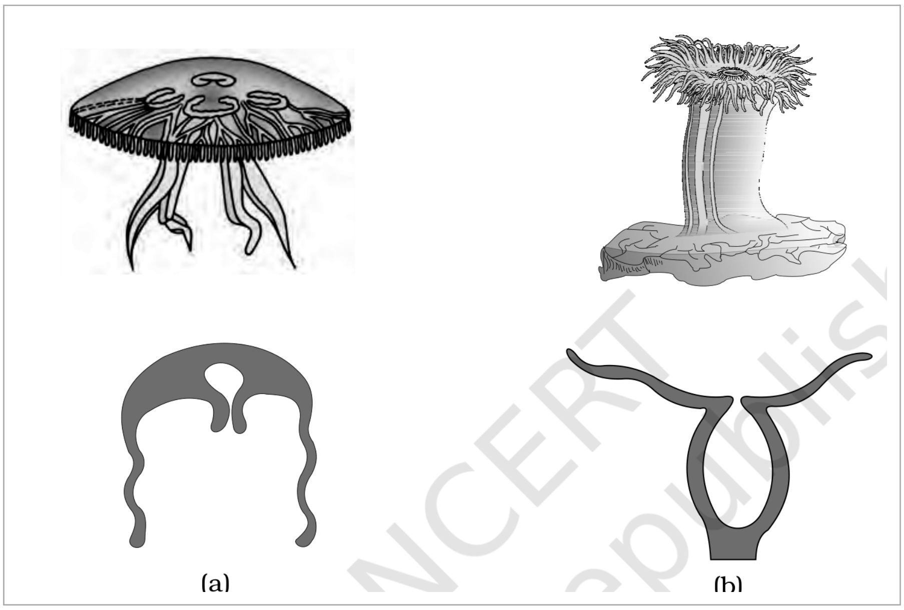

Fig. 4.6 *- Examples of Coelenterata indicating outline of their body form : (a) Aurelia (Medusa) (b) Adamsia (Polyp).*

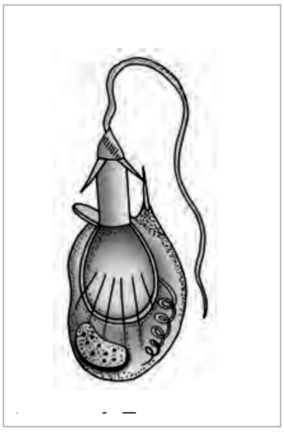

Fig. 4.7 *- Diagrammatic view of Cnidoblast.*

4.2.3 Phylum - Ctenophora

- Ctenophores, commonly known as **sea walnuts** or **comb jellies**, are exclusively marine, radially symmetrical, diploblastic organisms with tissue level of organisation. The body bears **eight external rows of ciliated comb plates**, which help in locomotion (Figure 4.8).
- Digestion is both extracellular and intracellular. **Bioluminescence** (the property of a living organism to emit light) is well-marked in ctenophores.
- Sexes are not separate. Reproduction takes place only by sexual means. Fertilisation is external with indirect development.
- **Examples:** *Pleurobrachia* and *Ctenoplana*.

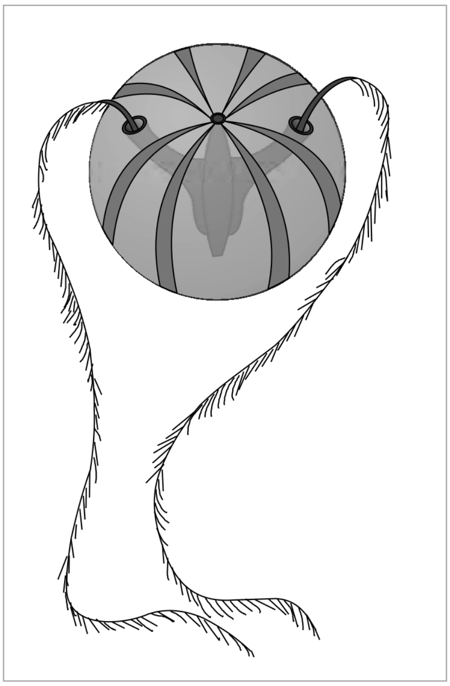

Fig. 4.8 *- Example of Ctenophora (Pleurobrachia).*

4.2.4 Phylum - Platyhelminthes

- They have a dorso-ventrally flattened body, hence are called **flatworms** (Figure 4.9). These are mostly **endoparasites** found in animals including human beings. Flatworms are bilaterally symmetrical, triploblastic and **acoelomate** animals with organ level of organisation.
- Hooks and suckers are present in the parasitic forms. Some of them absorb nutrients from the host directly through their body surface. Specialised cells called **flame cells** help in osmoregulation and excretion.
- Sexes are not separate. Fertilisation is internal and development is through many larval stages. Some members like *Planaria* possess high regeneration capacity.
- **Examples:** *Taenia* (Tapeworm), *Fasciola* (Liver fluke).

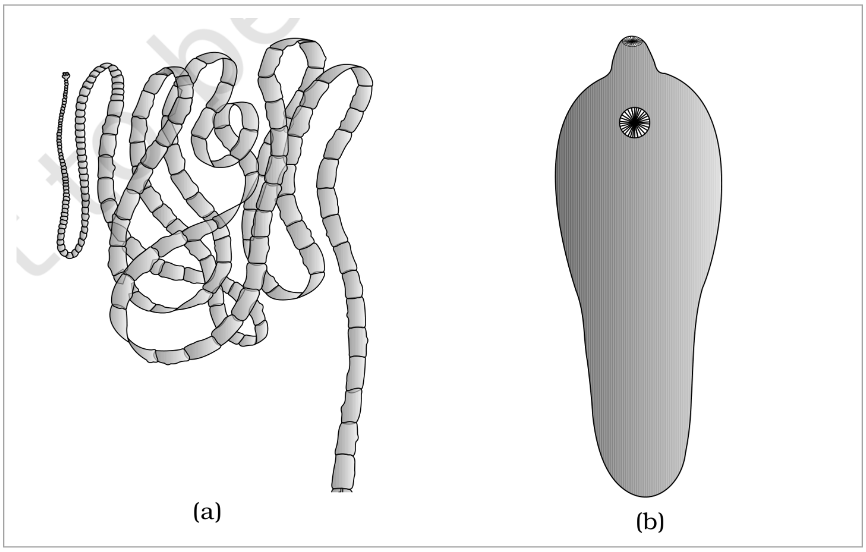

Fig. 4.9 *- Examples of Platyhelminthes : (a) Tape worm (b) Liver fluke.*

4.2.5 Phylum - Aschelminthes

- The body of the aschelminthes is circular in cross-section, hence the name **roundworms** (Figure 4.10). They may be free-living, aquatic and terrestrial or parasitic in plants and animals. Roundworms have **organ-system level** of body organisation.
- They are bilaterally symmetrical, triploblastic and **pseudocoelomate** animals. Alimentary canal is complete with a well-developed muscular pharynx. An excretory tube removes body wastes from the body cavity through the excretory pore.

- Sexes are separate (**dioecious**), i.e., males and females are distinct. Often females are longer than males. Fertilisation is internal and development may be direct (the young ones resemble the adult) or indirect.
- **Examples:** *Ascaris* (Roundworm), *Wuchereria* (Filaria worm), *Ancylostoma* (Hookworm).

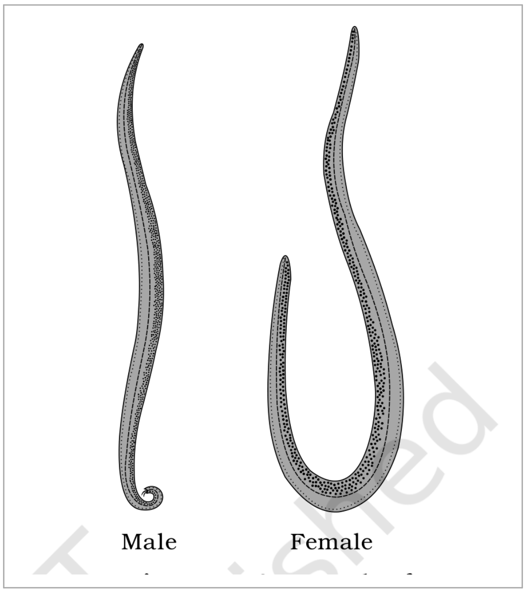

Fig. 4.10 *- Example of Aschelminthes : Roundworm. Labels: Male and Female.*

4.2.6 Phylum - Annelida

- They may be aquatic (marine and fresh water) or terrestrial; free-living, and sometimes parasitic. They exhibit organ-system level of body organisation and bilateral symmetry. They are triploblastic, **metamerically segmented** and coelomate animals.
- Their body surface is distinctly marked out into segments or **metameres** and, hence, the phylum name Annelida (Latin, *annulus* : little ring) (Figure 4.11). They possess longitudinal and circular muscles which help in locomotion. Aquatic annelids like *Nereis* possess lateral appendages, **parapodia**, which help in swimming.
- A **closed circulatory system** is present. **Nephridia** (sing. nephridium) help in osmoregulation and excretion. Neural system consists of paired ganglia (sing. ganglion) connected by lateral nerves to a double ventral nerve cord.
- *Nereis*, an aquatic form, is dioecious, but earthworms and leeches are monoecious. Reproduction is sexual.
- **Examples:** *Nereis*, *Pheretima* (Earthworm) and *Hirudinaria* (Blood sucking leech).

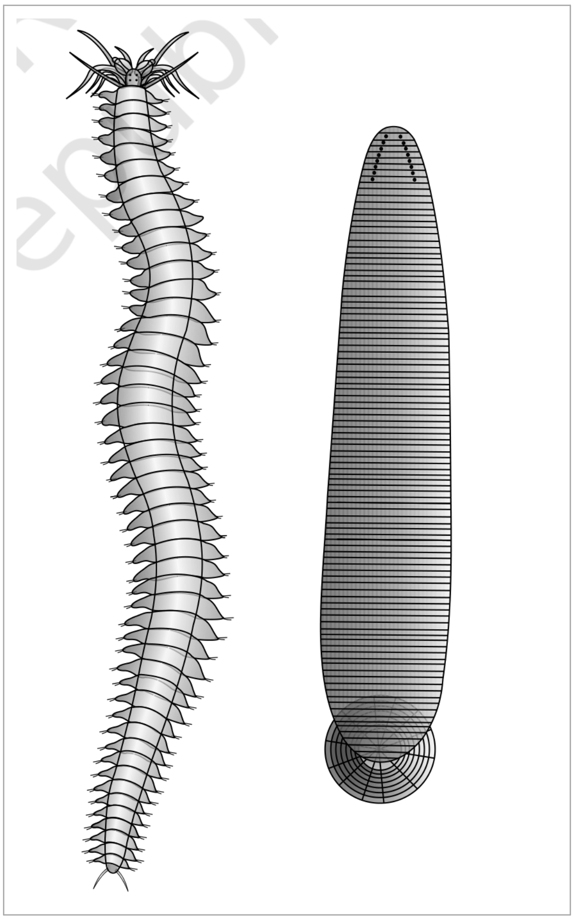

Fig. 4.11 - Examples of Annelida : (a) Nereis (b) Hirudinaria.

4.2.7 Phylum - Arthropoda

- This is the **largest phylum** of Animalia which includes insects. Over two-thirds of all named species on earth are arthropods (Figure 4.12). They have organ-system level of organisation.
- They are bilaterally symmetrical, triploblastic, segmented and coelomate animals. The body of arthropods is covered by **chitinous exoskeleton**. The body consists of **head, thorax and abdomen**. They have jointed appendages (*arthros*-joint, *poda*-appendages).
- Respiratory organs are gills, book gills, book lungs or tracheal system. Circulatory system is of **open type**. Sensory organs like antennae, eyes (compound and simple), statocysts or balancing organs are present. Excretion takes place through **malpighian tubules**.
- They are mostly dioecious. Fertilisation is usually internal. They are mostly oviparous. Development may be direct or indirect.
- **Examples:** Economically important insects - *Apis* (Honey bee), *Bombyx* (Silkworm), *Laccifer* (Lac insect); Vectors - *Anopheles*, *Culex* and *Aedes* (Mosquitoes); Gregarious pest - *Locusta* (Locust); Living fossil - *Limulus* (King crab).

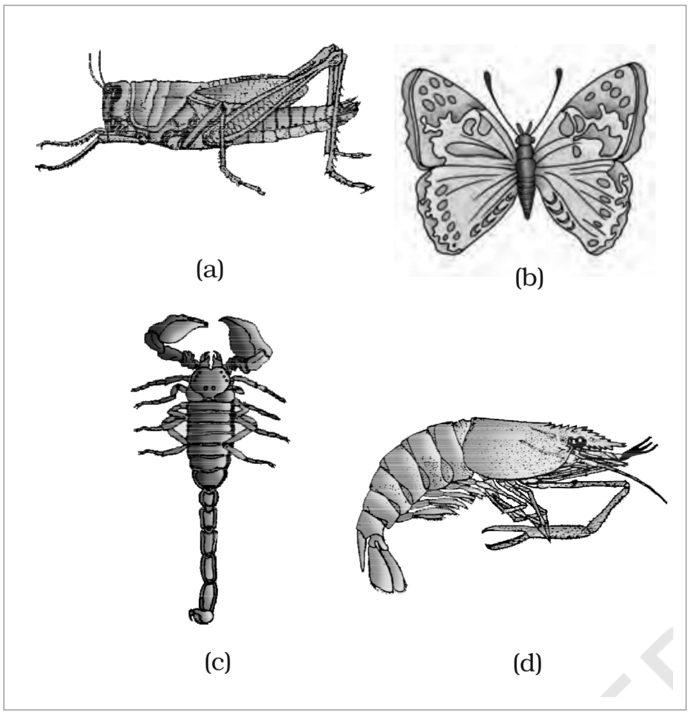

Fig. 4.12 *- Examples of Arthropoda : (a) Locust (b) Butterfly (c) Scorpion (d) Prawn.*

4.2.8 Phylum - Mollusca

- This is the **second largest animal phylum** (Figure 4.13). Molluscs are terrestrial or aquatic (marine or fresh water) having an organ-system level of organisation. They are bilaterally symmetrical, triploblastic and coelomate animals.
- Body is covered by a **calcareous shell** and is unsegmented with a distinct head, muscular foot and visceral hump. A soft and spongy layer of skin forms a **mantle** over the visceral hump. The space between the hump and the mantle is called the **mantle cavity** in which feather-like gills are present. They have respiratory and excretory functions.
- The anterior head region has sensory tentacles. The mouth contains a file-like rasping organ for feeding, called **radula**. They are usually dioecious and oviparous with indirect development.
- **Examples:** *Pila* (Apple snail), *Pinctada* (Pearl oyster), *Sepia* (Cuttlefish), *Loligo* (Squid), *Octopus* (Devil fish), *Aplysia* (Sea-hare), *Dentalium* (Tusk shell) and *Chaetopleura* (Chiton).

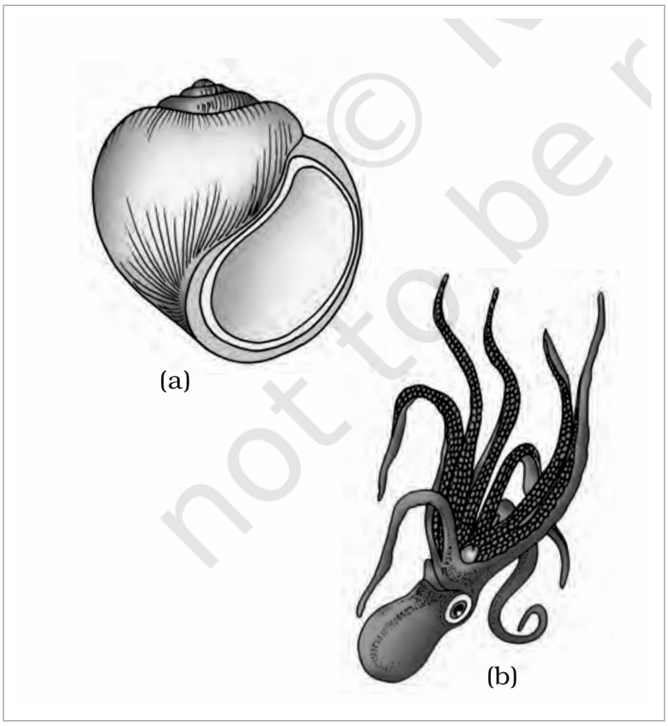

Fig. 4.13 *- Examples of Mollusca : (a) Pila (b) Octopus.*

4.2.9 Phylum - Echinodermata

- These animals have an endoskeleton of **calcareous ossicles** and, hence, the name Echinodermata (Spiny bodied, Figure 4.14). All are marine with organ-system level of organisation.
- The adult echinoderms are radially symmetrical but larvae are bilaterally symmetrical. They are triploblastic and coelomate animals. Digestive system is complete with mouth on the lower (ventral) side and anus on the upper (dorsal) side.
- The most distinctive feature of echinoderms is the presence of **water vascular system** which helps in locomotion, capture and transport of food and respiration. An excretory system is absent.
- Sexes are separate. Reproduction is sexual. Fertilisation is usually external. Development is indirect with free-swimming larva.
- **Examples:** *Asterias* (Star fish), *Echinus* (Sea urchin), *Antedon* (Sea lily), *Cucumaria* (Sea cucumber) and *Ophiura* (Brittle star).

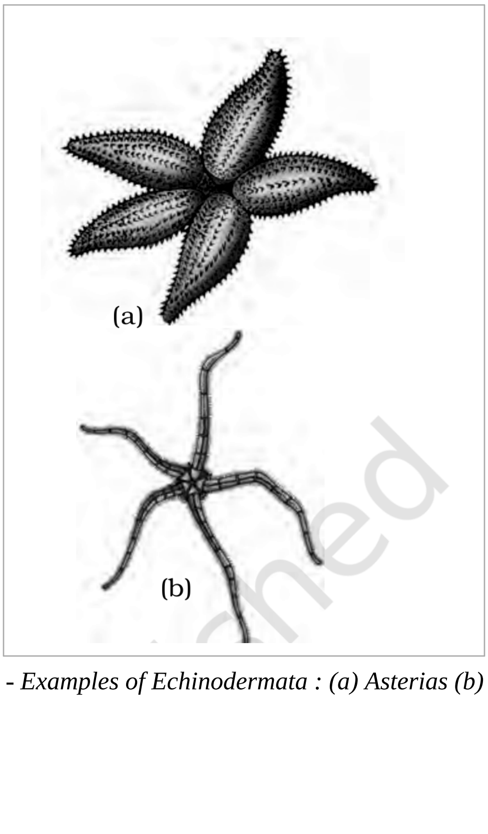

Fig. 4.14 *- Examples of Echinodermata : (a) Asterias (b) Ophiura.*

4.2.10 Phylum - Hemichordata

- Hemichordata was earlier considered as a sub-phylum under phylum Chordata. But now it is placed as a separate phylum under non-chordata. Hemichordates have a rudimentary structure in the collar region called **stomochord**, a structure similar to notochord.
- This phylum consists of a small group of worm-like marine animals with organ-system level of organisation. They are bilaterally symmetrical, triploblastic and coelomate animals. The body is cylindrical and is composed of an anterior **proboscis**, a **collar** and a long **trunk** (Figure 4.15).
- Circulatory system is of open type. Respiration takes place through gills. Excretory organ is **proboscis gland**. Sexes are separate. Fertilisation is external. Development is indirect.
- **Examples:** *Balanoglossus* and *Saccoglossus*.

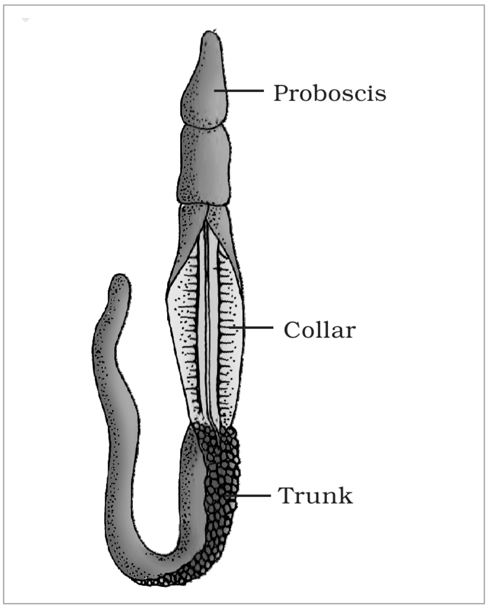

Fig. 4.15 *- Balanoglossus. Labels: Proboscis, Collar and Trunk.*